From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2014

FULL TEST – II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 09)** contains 09 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (10 – 13) contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (14 – 19) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (1 – 03)** contains 3 Matrix Match Type (4 × 4 Matrix) questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+6 marks** for all correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Two identical cylindrical vessels with their bases at the same level each contains a liquid of density ρ. The height of the liquid in one vessel is h_1 and that in other is h_2. The area of either base is A. Then the work done by gravity in equalizing the levels when the vessels are connected

(A) $\frac{1}{4}\rho gA\left(h_1-h_2\right)^2$ (B) $\frac{1}{4}\rho gA\left(h_1^2-h_2^2\right)$

(C) $\frac{1}{2}\rho gA\left(h_1-h_2\right)^2$ (D) $\frac{1}{2}\rho gA\left(h_1^2-h_2^2\right)$

2. An empty plastic box of mass m is found to accelerate up at a rate of g/6 when placed deep inside water. Then the mass of sand be filled in it so that it may accelerate down at a rate of g/6

(A) $\frac{5}{2}m$ (B) $\frac{2}{5}m$

(C) $\frac{2}{3}m$ (D) $\frac{3}{2}m$

3. The distance from the surface of the earth at which the acceleration due to gravity is the same below and above the surface of the earth

(A) $\left(\sqrt{5}+1\right)R_e$ (B) $\left(\frac{\sqrt{5}+1}{2}\right)R_e$

(C) $\frac{\left(\sqrt{5}-1\right)R_e}{2}$ (D) $\left(\sqrt{5}-1\right)R_e$

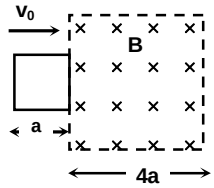
4. A ray is incident on a spherical body ($\mu = \sqrt{3}$) making an angle 60° with the normal drawn at the point of incidence. The ray after passing through sphere gets incident on the further surface of sphere and get reflected and refracted. Then the angle between the reflected and refracted rays at this surface is

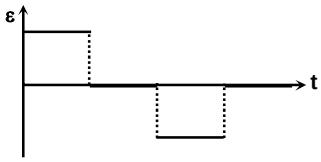
(A) 45° (B) 90°

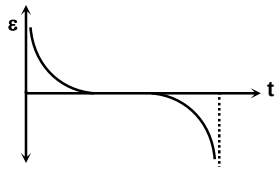
(C) 120° (D) 150°

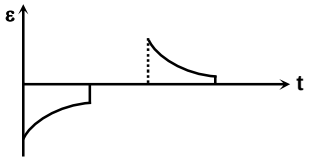
Space for rough work

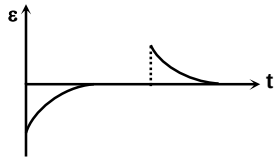
5. A square loop of side a is projected at t = 0, with speed v_0 into a region having uniform magnetic field B, with velocity perpendicular to the magnetic field. Which of the following plots correctly represents the variation of induced emf of loop with time



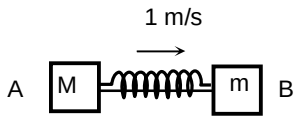
(A) 

(B) 

(C) 

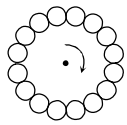
(D) 

6. A block A of mass M = 2kg is connected to another block B of mass 1kg with a string and a spring of force constant k = 600 N/m as shown in the figure. Initially spring is compressed to 10 cm and whole system is moving on a smooth surface with a velocity v = 1 m/s. At any time thread is burnt, the velocity of block A, when B is having maximum velocity w.r.t. ground, is:



(A) 0 (B) 1 m/s
(C) 3 m/s (D) none of these

7. A metallic chain with a length ℓ and whose ends are joined together is fitted onto a wooden disc as shown in the figure. The disc rotates with a speed of n revolutions per second. Then the tension of the chain T if its mass is m



(A) $m\ell n$ (B) $2m\ell n$
(C) $2m\ell n^2$ (D) $m\ell n^2$

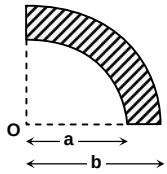
Space for rough work

8. An object of mass m is projected with momentum P at such an angle that its maximum height (H) is $1/4^{th}$ of its horizontal range (R). Its minimum kinetic energy in its path will be

(A) $\frac{p^2}{8m}$ (B) $\frac{p^2}{4m}$

(C) $\frac{3p^2}{4m}$ (D) $\frac{p^2}{m}$

9. The distance of centre of mass from O of a uniform plate having quarter circular inner and outer boundaries of radii a and b from centre O.



(A) $\frac{4}{3\pi}\left(\frac{b^3-a^3}{b^2-a^2}\right)$ (B) $\frac{4\sqrt{2}}{3\pi}\left(\frac{b^3+a^3}{b^2-a^2}\right)$

(C) $\frac{4}{\pi}\left(\frac{b^3+a^3}{b^2+a^2}\right)$ (D) $\frac{4\sqrt{2}}{3\pi}\left(\frac{b^3-a^3}{b^2-a^2}\right)$

Reasoning Type

This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

10. STATEMENT-1: In isothermal quasi-static process, heat is absorbed or given out by the system even though at every state the gas is at the same temperature as that of the surrounding reservoir.
and
STATEMENT-2: This is possible because of the infinitesimal difference in temperature between the system and surrounding.
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

11. STATEMENT-1: Only longitudinal waves can propagate in gases.
and
STATEMENT-2: The modulus of rigidity of gases is zero.
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

12. STATEMENT-1: If the net external force on a system is zero, the net torque on the system is independent of origin.
and
STATEMENT-2: When the net force on a system is zero, net torque may or may not be zero.
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

13. STATEMENT-1: In a photocell experiment, the stopping potential increases when the cathode is changed from cesium (Cs) metal to aluminium (Aℓ).
and
STATEMENT-2: Stopping potential does not depend on the threshold wavelength.
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

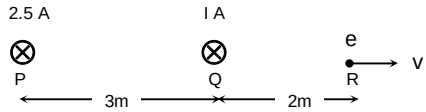
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Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 14 to 16

Two long parallel wires carrying currents 2.5 amp and I amp in the same direction as shown in figure. An electron moving with a velocity 4×10^5 m/sec. as shown in figure and experiences a force of magnitude 3.2×10^{-20} N at the point R. Then



14. The value of current I is
(A) 2 amp.
(B) 8 amp.
(C) 4 amp.
(D) 1 amp.

15. The position between PR at which a third long parallel wire carrying current 2.5 amp may be placed such that magnetic induction at R is zero.
(A) at 1 m from R and the direction of current is inward.
(B) at 1 m from R and the direction of current is outward
(C) at 1 m from P and the direction of current is inward.
(D) at 1 m from P and the direction of current is outward

16. The position not between PR but on the line joining PR at which a third long parallel wire carrying current 2.5 amp may be placed such that magnetic induction at R is zero.
(A) at a distance of 1m from R and the direction of current is outward.
(B) at a distance of 1 m from R and the direction of current is inward
(C) at a distance of 1 m from P and the direction of current is outward
(D) at a distance of 1 m from P and the direction of current is inward

Space for rough work

Paragraph for Question Nos. 17 to 19

A tube of length λ and radius R carries a steady flow of a fluid whose density is ρ and viscosity η. The velocity v of flow is given by $v = v_0[1 - (r^2/R^2)]$, where r is the distance of flowing fluid from the axis. Then

17. Volume of fluid flowing across the section of tube in unit time is

(A) $\frac{1}{4}\pi v_0 R^2$ (B) $\frac{1}{2}\pi v_0 R^2$

(C) $\pi v_0 R^2$ (D) $\frac{1}{8}\pi v_0 R^2$

18. Kinetic energy of the fluid within the volume of the tube is

(A) $\frac{1}{6}\pi \ell R^2 \rho v_0^2$ (B) $\frac{1}{8}\pi \ell R^2 \rho v_0^2$

(C) $\frac{1}{4}\pi \ell R^2 \rho v_0^2$ (D) $\frac{1}{3}\pi \ell R^2 \rho v_0^2$

19. The difference of pressure at the end of the tube is

(A) $\frac{4\eta \ell v_0}{R^2}$ (B) $\frac{8\eta \ell v_0}{R^2}$

(C) $\frac{2\eta \ell v_0}{R^2}$ (D) $\frac{\eta \ell v_0}{R^2}$

Space for rough work

SECTION – B

(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE or MORE is/are correct.**

1. Match the following:

Column – I	Column – II
(A) Linear combination of two SHM's	(p) $T = 2\pi\sqrt{\frac{R}{g}}$ and not $T = \infty$
(B) $y = A \sin \omega_1 t + A \sin (\omega_2 t + \phi)$	(q) SHM for equal frequencies and amplitude
(C) Time period of a pendulum of infinite length.	(r) Superposition may not be a SHM always.
(D) Maximum value of time period of an oscillating pendulum.	(s) amplitude will be $\sqrt{2}$ A for $\omega_1 = \omega_2$ and phase difference of $\pi/2$.

2. Match the following:

Column – I	Column – II
(A) Characteristic X-ray	(p) Inverse photoelectric effect.
(B) X-ray production	(q) high potential difference
(C) Cut-off wavelength	(r) Moseley's law
(D) Continues X-ray	(s) Brems-strahung radiation.

3. Considering a projectile motion.

Column – I	Column – II
(A) Change in magnitude of momentum will be present	(p) At highest point parabolic path
(B) Maximum angular momentum about the point of throw	(q) 2mv sin θ between the point of throw and target.
(C) minimum velocity	(r) $L = \frac{2mv^3 \sin^2 \theta \cos \theta}{g}$
(D) magnitude of change in momentum	(s) along vertical direction.

Space for rough work

Chemistry

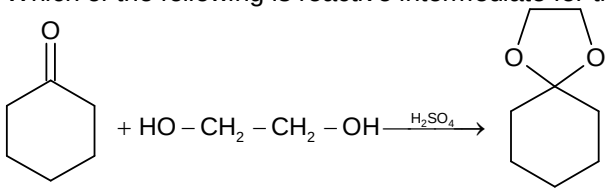
PART – II

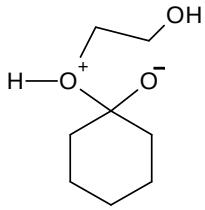
SECTION – A

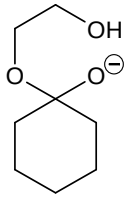
Straight Objective Type

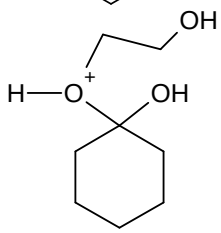
This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

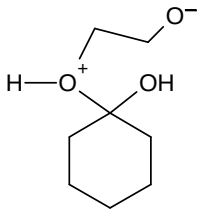
1. Which of the following is reactive intermediate for the following reaction?



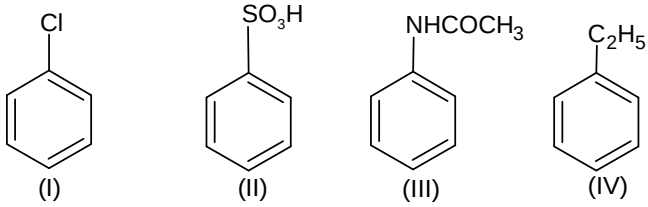
(A)


(B)


(C)


(D)


2. What is the expected order of reactivity of the following compounds in electrophilic nitration (more reactive, less reactive)?



(I) (II) (III) (IV)

(A) I > II > III > IV
(B) IV > I > II > III
(C) III > IV > I > II
(D) II > I > IV > III

Space for rough work

3. An electron is continuously accelerated in vacuum tube by applying potential difference. If its de-Broglie wavelength is decreased by 1%, the change in kinetic energy of the electron is nearly:
(A) decreased by 1.0% (B) increased by 2.0%
(C) increased by 1.0% (D) decreased by 2.0%

4. Quick lime; (CaO) is produced by heating limestone $CaCO_3$ to drive off CO_2 gas:
$CaCO_3 \longrightarrow CaO(s) + CO_2(g)$
$\Delta H = 180 \text{ kJ mol}^{-1}$
$\Delta S = 150 \text{ JK}^{-1}$
Assuming that variation of enthalpy change and entropy change with temperature to be negligible, which of the following is correct?
(A) Decomposition of $CaCO_3$(s) is always non-spontaneous at room temperature.
(B) Decomposition of $CaCO_3$(s) becomes spontaneous when temperature is less than 927^oC.
(C) Decomposition of $CaCO_3$(s) becomes non-spontaneous when temperature is greater than 1200^oC.
(D) All of the above.

5. A doctor by mistake administers a dilute $Ba(NO_3)_2$ solution to a patient for radiographic investigations. Which of the following should be the best to prevent the absorption of soluble barium and subsequent barium poisoning?
(A) NaCl (B) Na_2SO_4
(C) Na_2CO_3 (D) NH_4Cl

6. Match the products in the following reactions:

List – I	List – II
(1) $PCl_5 + H_2O \xrightarrow{1:1}$	(a) $N_3P_3Cl_6$ + 12 HCl
(2) $PCl_5 + H_2O(\text{excess}) \longrightarrow$	(b) $[PCl_4]^+ [AlCl_4]^-$
(3) $PCl_5 + AlCl_3 \longrightarrow$	(c) H_3PO_4 + HCl
(4) $PCl_5 + NH_4Cl \longrightarrow$	(d) $POCl_3$ + HCl

(A) 1-d, 2-c, 3-b, 4-a (B) 1-b, 2-b, 3-d, 4-a
(C) 1-d, 2-a, 3-b, 4-c (D) 1-c, 2-b, 3-d, 4-a

7. The magnetic momentum of $K[Fe(CN)_5(NO)]$ is:
(A) 1.73 BM (B) 2.84 BM
(C) 6.9 BM (D) Zero

Space for rough work

8. Glucose $\xrightarrow[(C_6H_5NHNH_2)]{\text{Phenylhydrazine}}$ X

Fructose $\xrightarrow{\text{Phenylhydrazine}}$ Y

X and Y are:

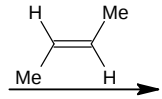
(A) Different osazones

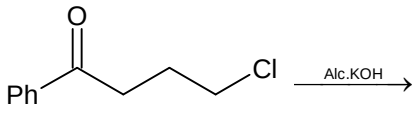
(B) Same osazones

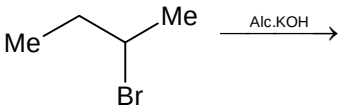
(C) Each molecule of glucose reacts with 4 molecules of phenylhydrazine

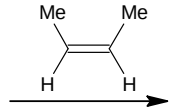
(D) Each molecule of fructose reacts with 2 molecules of phenylhydrazine

9. In which of the following α-elimination occurs:

(i) $Me_3CO^- + CHClBrI \longrightarrow$ 

(ii) 

(iii) 

(iv) $Ph_2CHCl + \overset{\ominus}{N}H_2 \longrightarrow$ 

(A) (i), (ii)

(B) (i), (iv)

(C) (ii), (iii)

(D) (ii), (iv)

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

10. STATEMENT-1: Bond energy of Cl – Cl bond is more than F – F bond.
and
STATEMENT-2: Shorter the bond length more is the bond energy.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

11. STATEMENT-1: A gas can be easily liquefied at any temperature below its critical temperature.
and
STATEMENT-2: Liquification of a gas takes place when the average kinetic energy of the molecules is low.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: Para methoxy aniline undergoes faster diazotization than aniline.
and
STATEMENT-2: Electron releasing group stabilizes diazonium salts.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

13. STATEMENT-1: The carboxylic and amino group in amino acid neutralize each other in aqueous solution.
and
STATEMENT-2: Amino acid forms Zwitterion by losing a proton from –COOH group and accepting a proton from H_2O molecule.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

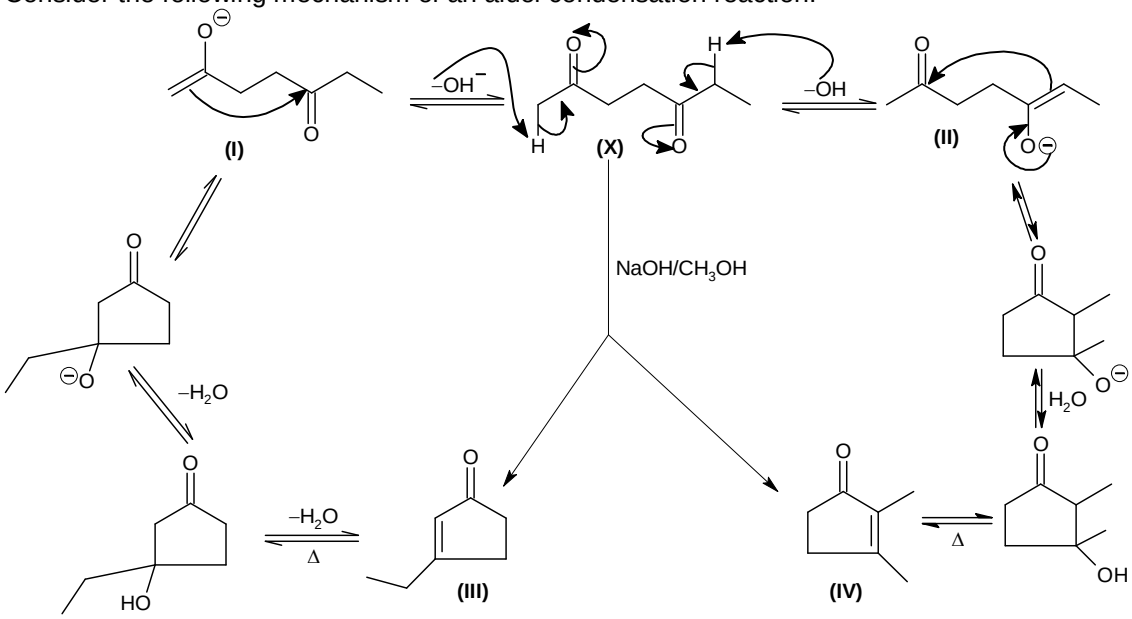
Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 14 to 16

Consider the following mechanism of an aldol condensation reaction:



Answer the following questions based on the above mechanism and your knowledge about aldol condensation.

14. Which of the following statement regarding enolates I and II is true?
    (A) Enolate I is more stable than II, since it is formed by deprotonation of more acidic hydrogen.
    (B) Enolate I is more stable than II, since it is less substituted.
    (C) Enolate II is more stable than I, since it is formed by deprotonation of more stable than acidic hydrogen.
    (D) Enolate II is more stable than I, since it is more substituted.

15. If all the alpha hydrogen in X is replaced by deuterium "D" atom. Which of the following would be observed? (Assuming high concentration of base)
    (A) Rate of reaction would decrease since C – D bond is stronger than C – H bond.
    (B) Rate of reaction would increase since D is more acidic than H.
    (C) Rate of reaction would remains same since C – D bond is not broken in the rate determining step.
    (D) III would be formed more easily than IV.

16. How many different aldol products (excluding stereoisomers) are expected from intermolecular aldol condensation of X?
    (A) 2 (B) 4
    (C) 6 (D) 8

Space for rough work

Paragraph for Question Nos. 17 to 19

The cell is $Mg|Mg^{+2}||Ag^{+}|Ag$

$E^{o}_{Ag^{+}|Ag} = 0.80V$

$E^{o}_{Mg|Mg^{2+}} = 2.37V$

17. The value of log K_c is:
(A) 107.45 (B) 80.5
(C) 67.15 (D) 53.725

18. The maximum work obtained by the cell is:
(A) 2×10^2 kJ (B) 3×10^2 kJ
(C) 5×10^2 kJ (D) 6×10^2 kJ

19. Calculate the E^o of cell.
(A) –3.17 V (B) 3.17 V
(C) 1.57 V (D) –1.57 V

Space for rough work

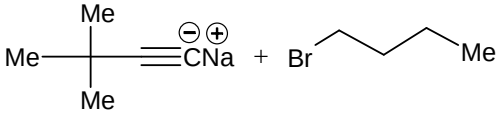
SECTION – B

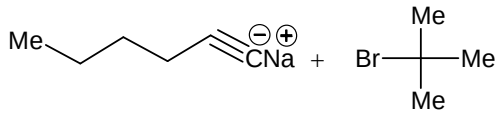
(Matching List Type)

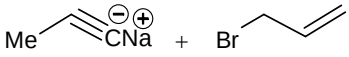
This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE or MORE is/are correct.**

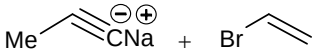
1. Match the Column – I with Column – II:

Column – I(Reaction)

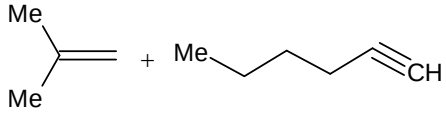
(A) 

(B) 

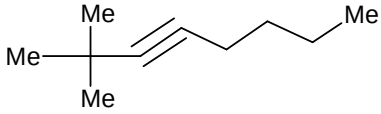
(C) 

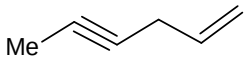
(D) 

Column – II(Products)

(p) 

(q) No reaction

(r) 

(s) 

2. Match the Column – I with Column – II:

Column – I(Orbitals)		**Column – II(Quantum No.)**	
(A)	2s	(p)	$n = 2,\ \ell = 0,\ m = 0$
(B)	$2p_z$	(q)	$n = 2,\ \ell = 1,\ m = 0$
(C)	$4d_{x^2-y^2}$	(r)	$n = 4,\ \ell = 1,\ m = \mp 2$
(D)	$4d_{z^2}$	(s)	$n = 4,\ \ell = 2,\ m = 0$

3. Match the Column – I with Column – II:

Column – I		**Column – II**	
(A)	NO_2	(p)	Paramagnetic
(B)	NO	(q)	Dimerises on cooling
(C)	Cl_2O	(r)	Disproportionate either in cold or hot water
(D)	ClO_2	(s)	Act both oxidizing and reducing agent

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Consider a set {1, 2, 3, ..., 100}. The number of ways in which a number can be selected from the set so that it is of the form x^y, (where x, y, $\in$ N and ≥ 2), is
   (A) 12 (B) 16
   (C) 5 (D) 11

2. Solution set of $[\sin^{-1}x] > [\cos^{-1}x]$, where [.] denotes greatest integer function
   (A) [sin 1, 1] (B) $\left[\frac{1}{\sqrt{2}}, 1\right]$
   (C) (cos 1, sin 1) (D) none of these

3. The equation $(1 + a^2)x^2 + 2a^2x + a^2 + b^2 - 1 = 0$ has roots of opposite sign if a + ib lies
   (A) on a straight line x + y = 1
   (B) inside a circle of centre (0, 0) and radius 1
   (C) on a parabola of vertex (0, 0) and focal length 1
   (D) none of these

4. The complex number z satisfying $|(z - 2 + 3i)| = 6$ and $|z - 4 - i| = |z - 12 - i|$ are
   (A) one (B) two
   (C) three (D) none of these

5. $f(x) = \left(x^2 - 1\right)^{20}\left(x^2 + x + 1\right)$, then
   (A) x = 1 is the point of local minima (B) x = –1/2 is the point of local minima
   (C) x = 1 is the point of local maxima (D) x = –1 is the point of local maxima

Space for rough work

6. If the tangent at the point (a sec α, b tan α) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}$ = 1 meets the transverse axis at T, then the distance of T from a focus of hyperbola is
(A) a(e $\pm$ cos α) (B) ae
(C) b(e + cos α) (D) $\sqrt{a^2e^2+b^2\cot^2\alpha}$

7. A bag contains 17 markers with numbers 1 to 17. A marker is drawn at random and then replaced; a second marker is drawn then the probability that first number is even and second is odd, is
(A) $\frac{72}{712}$ (B) $\frac{72}{289}$
(C) $\frac{9}{34}$ (D) none of these

8. If in a triangle ABC, $\sum \sin\frac{A}{2}=\frac{6}{5}$ and $\sum I\ \ I_1$ = 9 (where I_1, I_2, I_3 are ex–centres and I is incentre), then circumradius R is equal to
(A) $\frac{15}{8}$ (B) $\frac{15}{4}$
(C) $\frac{15}{2}$ (D) $\frac{4}{15}$

9. If $y^r = \frac{n\ !\ {}^{n+r-1}C_{r-1}}{r^n}$, where n = kr (k is constant), then $\lim_{n\to\infty} y$ is equal to
(A) (k + 1) log(1 + k) – k (B) (k + 1) log(k – 1) + k
(C) (k + 1) log(k – 1) – k (D) (k – 1) log(k – 1) + k

Reasoning Type

This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

10. STATEMENT 1: Locus of P(z) is a straight line in the equation $z - tz_1 = (1 - t)z_2$, where z_1, z_2 are the points A and B respectively in an argand plane
STATEMENT 2: A point dividing the two fixed point in the ratio m : n is collinear with the line through the given points
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

Space for rough work

11. STATEMENT 1: If $p < q$ and $px^2 + 4\mu xy + qy^2 + 4a(x + y + 1) = 0$ represents pair of straight lines for some $\mu \in R$ and $a \neq 0$, then $a \leq p$ or $a \geq q$
STATEMENT 2: The given equation has non-zero linear terms
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

12. STATEMENT 1: If $f(x) = ax^2 - bx + 2$; $a + b + 4 < 0$ then exactly one root lies between -1 and 0
STATEMENT 2: $ab < 0$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

13. STATEMENT 1: The order of the differential equation of family of circles touching a fixed straight line, which passes through origin is two
STATEMENT 2: General equation of circle has three parameters so if conditions are given then order is two always
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:
If (x_1, y_1) is the image (reflection) of the point (x, y) in the line $y = mx$, then

$$x_1 = \frac{1-m^2}{1+m^2}x + \frac{2m}{1+m^2}y \text{ and } y_1 = \frac{2m}{1+m^2}x + \frac{m^2-1}{1+m^2}y$$

$$\Rightarrow \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & \frac{m^2-1}{1+m^2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow X_1 = AX \Rightarrow X = A^{-1}X_1 \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & \frac{m^2-1}{1+m^2} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \text{ (since } A^2 = I).$$

14. The image of the parabola $y^2 = x$ in the line $x + y + 4 = 0$ is
(A) $x^2 = -(y + 4)$ (B) $(x + 4)^2 = -(y + 4)$
(C) $(x + 4)^2 = -y$ (D) none of these

Space for rough work

15. A ray moving along the line $px + qy = 1$, after striking a mirror placed along the line $y = 3$, is reflected along the line
(A) $px - qy = 1$ (B) $px - qy = 1 - q$
(C) $px - qy = 1 - 6q$ (D) $px - qy = 1 - 4q$

16. A tangent is drawn to the circle $x^2 + y^2 + 4x + 6y - 7 = 0$ at the point $(2, -1)$. The image of the tangent in the line $y = x + 2$ touches the parabola
(A) $y^2 = 5x$ (B) $5y^2 = x$
(C) $y^2 = -5x$ (D) $5y^2 = -x$

Paragraph for Question Nos. 17 to 19

Read the following write up carefully and answer the following questions:
Consider a three dimensional Cartesian system with origin at O and three rectangular coordinate axes x, y and z-axis. Suppose that the distance between two points P and Q in the space having their coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively be defined by the following formula
$d(P, Q) = |x_2 - x_1| + |y_2 - y_1| + |z_2 - z_1|$

Although the formula of distance between two points has been defined in a new way, yet the other definitions remain same (like section formula, direction cosines etc.). So, in general equations of straight line in space, plane in space remain unchanged. Now answer the following questions.

17. If l, m, n represent direction cosines (if we can call it) of a vector $\overrightarrow{OP}$, then which of the following relations holds?
(A) $l^2 + m^2 + n^2 = 1$ (B) $l + m + n = 1$
(C) $|l + m + n| = 1$ (D) $|l| + |m| + |n| = 1$

18. Locus of point P if $d(O, P) = k$, where k is a positive constant number, represents
(A) a sphere of radius k
(B) a set of eight planes forming an octahedron
(C) a set of eight planes forming hexagonal prism
(D) an infinite cylinder of radius k

19. Let OABC represents a regular tetrahedron with edge length 2a and vertices A, B, C lying on x, y and z-axis respectively. Let P be a point inside the region of the tetrahedron satisfying $d(O, P) \le \min\{d(P, A), d(P, B), d(P, C)\}$. The maximum value of d(O, P) is
(A) a (B) $\sqrt{2}$ a
(C) $\frac{2a}{3}$ (D) $\frac{4a}{3}$

Space for rough work

SECTION – B

(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE or MORE is/are correct.**

1. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	$\int \frac{dx}{(a^2-x^2)^2} - \frac{1}{2a^2}\int \frac{dx}{a^2-x^2}$ is equal to	(p)	$\frac{x}{2a^2(a^2-x^2)}+c$
(B)	$\int \frac{dx}{x^2\sqrt{x^2-a^2}}$	(q)	$\frac{\sqrt{x^2-a^2}}{a^2x}+c$
(C)	$\int \frac{\sqrt{a^2-x^2}}{x^2}dx$	(r)	$-\frac{\sqrt{a^2-x^2}}{x}+\cos^{-1}\frac{x}{a}+c$
(D)	$\int \frac{\sqrt{x^2-a^2}}{x}dx$	(s)	$\sqrt{x^2-a^2}-a\sec^{-1}\frac{x}{a}+c$

2. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	Area bounded by the parabola $y^2 = 4x$ and the line $y = x$ is	(p)	$\frac{1}{2}$
(B)	$\int_0^{\pi/2} (\sin^5 x + \cos^5 x)dx$ is equal to	(q)	–22
(C)	If S = 0 be the equation of the hyperbola $x^2 + 4xy + 3y^2 - 4x + 2y + 1 = 0$, then the value of k for which S + k = 0 represents its asymptotes is	(r)	$\frac{8}{3}$
(D)	The value of α for which $\alpha\hat{i}+2\hat{j}+\hat{k}$, lies in the plane containing three points $\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}+2\hat{j}+\hat{k}$ and $3\hat{i}-\hat{k}$ is	(s)	$\frac{16}{15}$

3. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	$3x^2 + 2(b^2 + 1)x + b^2 - 3b + 2 = 0$ will have roots of opposite sign if $b \in$	(p)	$\left(\frac{2}{3}, 1\right)$
(B)	The probability of a problem being solved by 3 students are $\frac{1}{2}, \frac{1}{3}$ and $\alpha \in (0, 1)$. Then probability that problem will be solved $\in$	(q)	$[2\sqrt{2}, \infty)$
(C)	If 3 real numbers satisfy the equation $x + y + z = 5$ and $xy + yz + zx = 8$, then $z \in$	(r)	$\left[1, \frac{7}{3}\right]$
(D)	The range of $\lvert\sec x\rvert + \lvert\operatorname{cosec} x\rvert$ is	(s)	(1, 2)

Space for rough work